

# Employee Salary Prediction System

Machine Learning & Data Analytics Project



Predicting Employee Salaries  
using **Python** & **Machine Learning**



Developed By:  
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Technologies Used:  
**Python** | **Pandas** | **NumPy** | **Scikit-Learn** | **Matplotlib**



[illegible]

## Importing Required Libraries

```
[5]  
✓ 0s  import pandas as pd  
import numpy as np  
from sklearn.linear_model import LinearRegression
```

## Loading the Dataset

```
[6]  
✓ 0s  df = pd.read_csv("/content/employee_salary_prediction_utf8.csv")
```



## Displaying First 5 Rows of Dataset

This step is used to preview the dataset and understand the structure of the data. It helps in checking column names, values, and overall dataset format before analysis.

[4]

✓ 0s



```
df.head()
```

...

|  | Employee_ID | Age | YearsExperience | Department | Salary |
|--|-------------|-----|-----------------|------------|--------|
|--|-------------|-----|-----------------|------------|--------|



|   |   |    |      |       |        |
|---|---|----|------|-------|--------|
| 0 | 1 | 29 | 9.5  | Sales | 93382  |
| 1 | 2 | 37 | 5.6  | Sales | 71700  |
| 2 | 3 | 44 | 11.0 | HR    | 108807 |
| 3 | 4 | 38 | 3.8  | Sales | 55938  |
| 4 | 5 | 30 | 7.6  | IT    | 85156  |



## Dataset Information

In this step, I checked the dataset structure, data types, column names, and non-null values to better understand the data before preprocessing.

[5]

✓ 0s



df.info()



```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 5 columns):
#   Column             Non-Null Count  Dtype
---  -
0   Employee_ID        100 non-null   int64
1   Age                 100 non-null   int64
2   YearsExperience     100 non-null   float64
3   Department          100 non-null   object
4   Salary              100 non-null   int64
dtypes: float64(1), int64(3), object(1)
memory usage: 4.0+ KB
```



## Statistical Summary

This step provides statistical insights such as mean, median, minimum, maximum, and standard deviation of numerical columns in the dataset.

[6]

✓ 0s



```
df.describe()
```



|       | Employee_ID | Age        | YearsExperience | Salary        |
|-------|-------------|------------|-----------------|---------------|
| count | 100.000000  | 100.000000 | 100.000000      | 100.000000    |
| mean  | 50.500000   | 35.650000  | 8.529000        | 89504.840000  |
| std   | 29.011492   | 8.398382   | 3.824913        | 26792.120221  |
| min   | 1.000000    | 21.000000  | 1.200000        | 40058.000000  |
| 25%   | 25.750000   | 28.750000  | 5.575000        | 70542.250000  |
| 50%   | 50.500000   | 35.000000  | 8.100000        | 85867.000000  |
| 75%   | 75.250000   | 42.250000  | 11.850000       | 112285.750000 |
| max   | 100.000000  | 50.000000  | 15.000000       | 137915.000000 |





## Checking Missing Values

In this step, I checked the dataset for missing or null values to identify incomplete data entries before preprocessing. `df.isnull().sum()`

[7]

✓ 0s



```
df.isnull().sum()
```

...

0

Employee\_ID 0

Age 0

YearsExperience 0

Department 0

Salary 0

dtype: int64



## Removing Missing Values

This step removes null values from the dataset to improve data quality and ensure accurate machine learning model performance. `df.dropna(inplace=True)`

[8]

✓ 0s



```
df.dropna(inplace=True)
```



## Selecting Features for Prediction

In this step, I selected important input features such as Age and Years of Experience to train the machine learning model for salary prediction.

[21]

✓ 0s



```
X = df[["Age", "YearsExperience"]]
```

## Defining Target Variable

In this step, I selected the target variable (Salary) that the machine learning model will predict based on the input features.

```
[32] ✓ 0s y = df["Salary"]
```

## Initializing Linear Regression Model

In this step, I initialized the Linear Regression model, which is used to predict employee salaries based on input features.

```
[33] ✓ 0s model = LinearRegression()
```

## Training the Machine Learning Model

In this step, I trained the Linear Regression model using the selected features and target variable to learn patterns from the dataset.

```
[34] ✓ 0s model.fit(X, y)
```

▼ ...

- LinearRegression ⓘ ?
- LinearRegression()

## Taking User Input

In this step, I collected user input such as age and years of experience to predict the estimated employee salary using the trained machine learning model.

```
[35] ✓ 20s age = float(input("Enter Your Age: "))
YearsExperience = float(input("Enter Experience in Years (6 months = 0.5): "))
```

▼

```
Enter Your Age: 25
Enter Experience in Years (6 months = 0.5): 3
```

## Predicting Employee Salary

In this step, I used the trained machine learning model to predict the employee salary based on age and years of experience provided by the user.

```
[37] ✓ 0s ▶ prediction = model.predict([[age, YearsExperience]])
```

## ✓ Final Output

In the final step, the machine learning model successfully predicted the employee salary based on the user's age and years of experience. The model estimated an approximate salary of ₹50,840, demonstrating how machine learning can be used for real-world salary prediction and data-driven decision making.

```
[38] ✓ 0s ▶ print("\nPredicted Salary = ₹", round(prediction[0]))  
...  
Predicted Salary = ₹ 50840
```

## **Project Conclusion**

The **Employee Salary Prediction System** was successfully developed using Machine Learning and Data Analytics techniques. The model analyzed employee age and years of experience to generate an estimated salary prediction of **₹50,840**.

### **This project provided hands-on experience in:**

- ✓ Data Cleaning & Preprocessing
- ✓ Exploratory Data Analysis (EDA)
- ✓ Machine Learning Model Training
- ✓ Predictive Analytics using Python

### **Technologies Used:**

Python | Pandas | NumPy | Scikit-Learn | Matplotlib